Studying the Impact of Post-Training Quantization on Fairness in Transformer Models *

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Abstract

We propose to study how post-training quantization (PTQ) affects fairness and internal representations in Transformer-based language models. Using a pretrained BERT encoder kept frozen, and a lightweight classifier trained on top, we will apply multiple numerical-precision settings (FP32 baseline, FP16, and INT8 PTQ) and measure both classification performance and fairness-related behavior. Fairness evaluation will combine group-level metrics (demographic parity, equal opportunity), counterfactual consistency tests (prediction flip-rate under demographic swaps), and representation-level analyses (layerwise L2 / cosine shifts and Centered Kernel Alignment (CKA)). The study aims to answer whether and how PTQ alters demographic sensitivity when the encoder is not fine-tuned, and to identify where in the model representation shifts (if any) are concentrated.

1 Introduction

Transformer-based pretrained language models have become central to modern NLP systems. To deploy these models efficiently, practitioners often apply post-training quantization (PTQ) to reduce memory and compute. While many works have shown that carefully applied quantization preserves aggregate performance (e.g., accuracy or F1) for a range of tasks, less is known about how quantization affects fairness: might reducing numerical precision change how models treat demographic attributes, or alter their internal representations in ways that increase demographic disparities?

This project isolates the effect of quantization by (1) using a pretrained BERT encoder as a frozen feature extractor, and (2) varying only the numerical precision of the encoder while keeping the (lightweight) classifier training in full precision. We will evaluate not only standard task metrics but also counterfactual and representation-level measures (including CKA) to obtain a layered view of how quantization perturbs model behavior.

Finally, to study the architecture-dependence of quantization-induced effects, we will extend our analysis beyond encoder-only BERT models to include decoder-only transformer architectures.

2 Background and Related Work

- **LLM Quantization for Code and Extreme Low-bit Regimes:** Giagnorio (2025) shows that code-generation LLMs up to 34B parameters can be quantized to 4 bits with minimal performance loss using code-specific calibration. While the study emphasizes quantization quality and calibration strategies, it does not address fairness, which is the focus of our work (Giagnorio, 2025).

- **Quantization and Fairness (Vision Models):** Liu et al. (Liu, 2025) study the fairness impact of quantization in vision models, finding that PTQ often worsens accuracy disparities across demographic groups. They further show that Quantization-Aware Training (QAT) can amplify these gaps even more than PTQ, while simple data augmentation or rebalancing strategies can partially mitigate the resulting unfairness.

- **Quantization and Fairness in Language Models:** Proskurina et al. (Proskurina et al., 2025) show that common quantization pipelines (e.g., GPTQ-style algorithms) can increase biased outputs and degrade fairness-benchmark performance for generative LLMs. They propose *Fair-GPTQ*, which incorporates explicit group-fairness constraints into the quantization objective to guide rounding and reduce stereotyped generation; their approach

*Code available at GitHub repository.

reduces group bias while keeping most of the efficiency benefits of low-bit quantization.

3 Objective and Motivation

We aim to determine whether post-training quantization inadvertently alters a language model's demographic sensitivity. Specifically, we will:

- Isolate the effect of quantization by using a pretrained BERT encoder as a fixed (frozen) feature extractor and training only a lightweight classifier on top. This avoids confounding fine-tuning updates.

- Compare models under different numerical precisions (e.g. FP32 baseline, FP16, INT8 PTQ) to see how accuracy and fairness change. Prior work suggests quantization can worsen group disparities, but it's unknown if a frozen-model setup exhibits this.

- Measure classification performance (accuracy, F1, etc.) in each case, and also evaluate fairness by subjecting test inputs to demographic perturbations (e.g. swapping names or identity terms). We will track how often the model's prediction changes under such counterfactual swaps.

- Analyze internal representations: we will extract hidden embeddings and logits at each layer for a set of inputs and quantify how they shift between precisions. This will reveal where quantization injects noise or bias into the model's internal state.

4 Datasets

We plan to utilize three complementary datasets to evaluate fairness at both the group and individual levels. Each dataset serves a distinct role in isolating the effects of quantization on model representations.

- **Jigsaw Unintended Bias in Toxicity Classification:** To evaluate group-level fairness metrics (Demographic Parity and Equal Opportunity), we use the Jigsaw dataset (Borkan et al., 2019). This corpus contains approximately 2 million user comments labeled for toxicity, with a subset explicitly tagged with identity labels (e.g., race, gender, religion). By treating this as a binary classification task (Toxic vs. Non-Toxic), we can measure if quantization disproportionately degrades performance for specific demographic subgroups.

- **Bias in Bios:** To evaluate the impact of quantization on stereotypical associations, we use the Bias in Bios dataset (De-Arteaga et al., 2019). This dataset consists of biographical texts annotated with one of 28 occupations (e.g., Professor, Physician, Nurse). It is widely used to study gender bias, as many occupations exhibit strong gender imbalances. We use this to measure *Equal Opportunity Difference* by comparing the True Positive Rates (TPR) for male and female instances within the same occupation before and after quantization.

- **Equity Evaluation Corpus (EEC):** To strictly measure counterfactual consistency, we employ the Equity Evaluation Corpus (Kiritchenko and Mohammad, 2018). EEC consists of 8,640 synthetic sentences generated from templates that vary only by demographic identity terms (e.g., names associated with different races or genders) while keeping the semantic context fixed. This dataset serves as a controlled test set (paired with a sentiment classifier trained on standard data like IMDB) to calculate the *Counterfactual Flip Rate* and measure representational shifts (CKA, Cosine Similarity) caused solely by numerical precision reduction.

5 Proposed Methodology

- **Model Setup:** Start with a pretrained BERT-base model (e.g. uncased). Keep all weights fixed to represent the frozen encoder. Attach a shallow classifier (e.g. one or two-layer MLP) on top of the [CLS] embedding for the target task. Train only this classifier (in full precision) on the task data.

- **Quantization:** Apply post-training quantization to the frozen BERT model. We will consider multiple schemes: full-precision FP32 (baseline), half-precision FP16, and 8-bit integers (INT8) via PTQ. For INT8 we will use a standard dynamic/static quantization toolkit (e.g. PyTorch PTQ or NVIDIA's TensorRT) as in prior work. We do *not* fine-tune BERT after quantization.

- **Classification & Fairness Testing:** For each precision setting (FP32/FP16/INT8), run the model+classifier on the test set. Record standard performance metrics (accuracy, F1, etc.). To evaluate fairness, for each test sentence we construct a parallel sentence with a demographic swap (e.g. changing "She" to "He", or "John" to "Jane"). We then check if the model's predicted label changes between the original and swapped sentences. We quantify this "flip rate" as well as group-wise prediction rates.

- **Representation Analysis:** Collect hidden representations (e.g. the [CLS] vector) from each BERT layer for a sample of inputs. Compare these vectors between FP32 and quantized models using metrics like cosine similarity and L2 distance.

  For example, for layer l, we compute:

$$\Delta_l^{(2)} = \frac{1}{N} \sum_{i=1}^{N} \|h_l^{\text{FP32}}(x_i) - h_l^{\text{Quant}}(x_i)\|_2,$$

$$\cos_l = \frac{1}{N} \sum_{i=1}^{N} \frac{\langle h_l^{\text{FP32}}(x_i), h_l^{\text{Quant}}(x_i) \rangle}{\|h_l^{\text{FP32}}(x_i)\|_2 \cdot \|h_l^{\text{Quant}}(x_i)\|_2}.$$

  Additionally, we will use **Centered Kernel Alignment (CKA)** to quantify the similarity between hidden representations across layers and quantization levels. CKA is invariant to orthogonal transformations and isotropic scaling, and is well-suited for comparing representational structures.

  Given two representation matrices X and Y of shape $N \times d$ (for N inputs), we compute linear CKA as:

$$\text{CKA}(X, Y) = \frac{\|X^\top Y\|_F^2}{\|X^\top X\|_F \cdot \|Y^\top Y\|_F},$$

  where $\|\cdot\|_F$ is the Frobenius norm. Higher CKA scores indicate greater alignment between FP32 and quantized representations. We will compute CKA per-layer and visualize alignment trends across depth and quantization settings.

- **Implementation Details:** We will follow standard PTQ procedures (calibrating with a held-out set if needed). Classifier training will be repeated with different random seeds to assess stability. Representations for analysis will be collected from a fixed sample of inputs to ensure consistency across quantization levels.

6 Evaluation

- **Performance Metrics:** Report classification accuracy and F1-score for each precision level (FP32, FP16, INT8) on held-out test data. This checks that quantization does not catastrophically degrade performance.

- **Fairness Metrics:** We measure two standard group fairness metrics:

  - **Demographic Parity Difference:**

$$\left| P(\hat{Y} = 1 \mid A = 0) - P(\hat{Y} = 1 \mid A = 1) \right|$$

  - **Equal Opportunity Difference:**

$$|\text{TPR}_{A=0} - \text{TPR}_{A=1}|$$

  - **Counterfactual Flip Rate:** the fraction of predictions that change under a demographic token swap:

$$\frac{1}{N} \sum_{i=1}^{N} \mathbb{I}\left(\hat{y}(x_i) \neq \hat{y}(x_i^{\text{swap}})\right)$$

- **Representation Shift:** For each layer l, compute the average L2 shift and cosine similarity (as defined above) between FP32 and quantized hidden states. Also compute layerwise CKA to measure representational alignment. Plot these differences to observe where quantization injects the most distortion. We expect deeper layers to accumulate more quantization noise.

- **Analysis:** We will compare how all metrics change from FP32 to FP16 to INT8. For example, do we see a sharp increase in demographic disparity at INT8 relative to FP16? We will analyze whether observed fairness degradations (if any) correlate with representational drift, group imbalance, or uncertainty.

- **Baselines and Ablations:** If fairness drops dramatically, we can consider simple mitigations (e.g. re-balancing data) to see if the effect is due to imbalance. A baseline with no sensitive attribute swaps will show aggregate bias metrics, and our counterfactual tests reveal hidden effects masked by aggregates.

7 Progress so Far

1. Comprehensive Data Collection

We have successfully secured and pre-processed three primary datasets, each serving a specialized role in identifying different facets of algorithmic bias:

- **Jigsaw Unintended Bias in Toxicity Classification:** We acquired this corpus to evaluate group-level fairness. The raw toxicity scores were binarized using a strict ≥ 0.5 threshold to establish ground-truth labels for "Toxic" vs "Non-Toxic" classification. We isolated identity attributes such as race, religion, and gender to calculate parity metrics.

- **Bias in Bios:** This professional corpus was integrated to analyze stereotypical associations in occupation prediction. It allows us to measure the True Positive Rate (TPR) disparities between male and female instances within 28 distinct professions.

- **Equity Evaluation Corpus (EEC):** We secured this synthetic dataset to serve as a controlled environment for counterfactual probing. It consists of 8,640 sentences where semantic context is held constant while demographic terms are swapped, allowing us to calculate the Counterfactual Flip Rate.

7.1 Model Architecture and Independent Baseline Training

We instantiated a `bert-base-uncased` architecture and explicitly froze all 110 million parameters by disabling computational graph tracking (setting `requires_grad=False`). This ensures that any observed bias stems strictly from numerical precision reduction rather than fine-tuning updates.

From this base, two completely separate model pipelines were trained:

- **Jigsaw Toxicity Baseline (FP32):** A lightweight feed-forward classification head was appended to the frozen BERT embeddings and trained exclusively on the *Jigsaw Toxicity* dataset for binary toxicity detection. A class-weighted loss function was necessary to mitigate the extreme imbalance between benign and toxic comments.

- **Bias in Bios Occupation Baseline (FP32):** A second, independent classification head was attached to the same frozen encoder architecture and optimized over **28 output logits** to predict professional occupations based on the biographical data.

Upon convergence of both models, we persisted the full-precision weights and extracted hidden state tensors for a deterministic sample of 1,000 inputs to serve as the geometric baseline for representational alignment.

4. Quantization Intervention and Metrics Computation

Precision Stratification: We utilized the `torch.quantization.quantize_dynamic` API to generate a compressed INT8 variant of the encoder. Additionally, we created an intermediate FP16 baseline using the `.half()` method to track the trajectory of fairness degradation.

Fairness Benchmarking: Using the `fairlearn` library, we computed:

- **Demographic Parity Difference (DPD)** to measure selection rate equality.

- **Equal Opportunity Difference (EOD)** to assess True Positive Rate disparities.

- **Counterfactual Flip Rate (CFR)** to identify prediction instability triggered by individual name or pronoun swaps.

5. Representation-Level Interrogation

Geometric Divergence: We executed a mathematical analysis script to compute L2 Distance and Cosine Similarity shifts across all 12 transformer layers.

Structural Alignment: We applied Centered Kernel Alignment (CKA) to map the macroscopic similarity of representational manifolds between FP32 and INT8 states. This allowed us to pinpoint specific layers where "representational collapse" occurs, correlating structural noise with the observed spikes in demographic bias.

8 Results (So Far...)

We successfully implemented a comparative analysis pipeline that evaluates a full-precision FP32 BERT baseline against FP16 and INT8 quantized variants to measure both performance and fairness trade-offs. By integrating the Fairlearn library

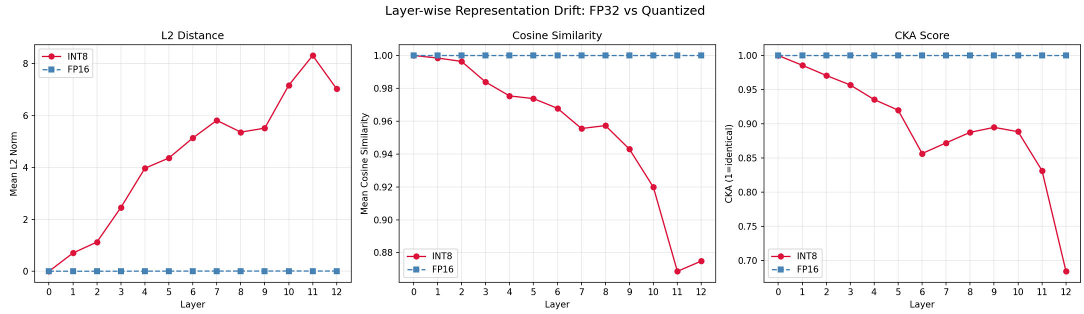


Figure 1: Layer-wise representation drift between FP32 and quantized models across transformer layers.

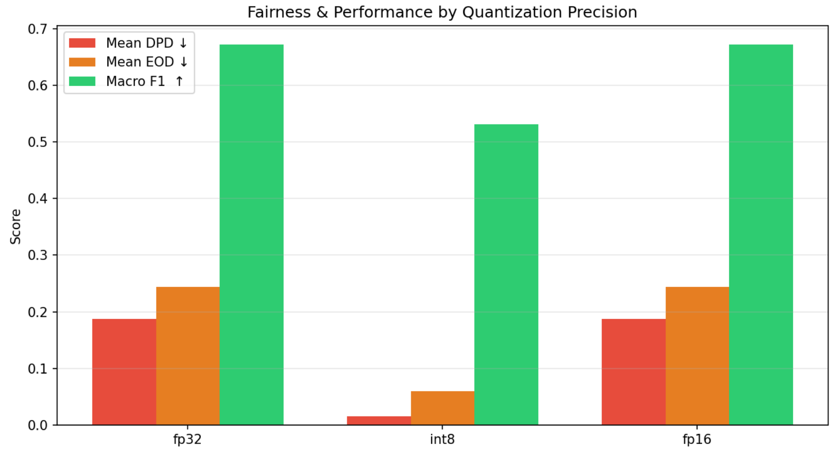


Figure 2: Fairness & Performance by Quantization Precision

and Centered Kernel Alignment (CKA), we quantified how reducing numerical precision induces representational drift and impacts group-level fairness metrics like Demographic Parity Difference and Equal Opportunity Difference across multiple identity attributes. Our layer-wise analysis demonstrated that quantization noise accumulates in deeper layers, allowing us to correlate structural manifold collapse with measurable spikes in algorithmic bias.

1. Linear Probing BERT-Uncased for Jigsaw Unintended Bias

1. Layer-wise Representation Drift (FP32 vs. Quantized)

Figure 1 This analysis tracks how internal representations deviate from the full-precision baseline as information propagates through the 12 transformer layers.

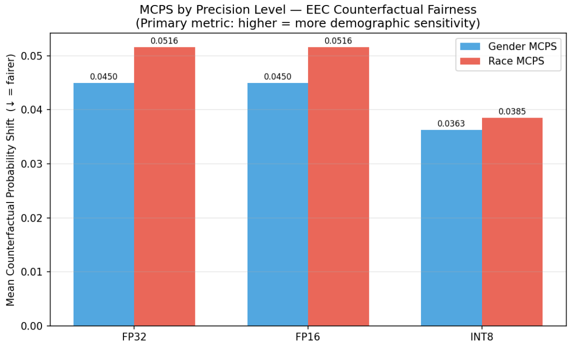
- **L2 Distance:** We observed a consistent increase in the L2 distance as the layer depth increases, this indicates the quantization noise is getting accumulated and this causing the physical disposition of embedding vectors further from their original position

- **Cosine Similarity:**The above observation is preserved here, as we moved deep the cosine similarity also dropped indicating that aggresive rounding fundamentally alters the contextual meaning captured by the encoder.

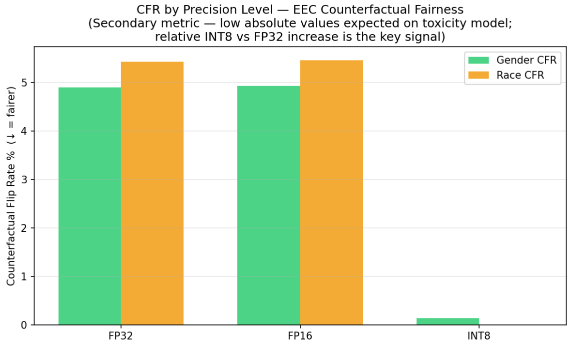
- **CKA Score:** The **Centered Kernel Alignment(CKA)** is our primary metric for structural manifold integrity,As we can see there is a significant collapse in the final encoder layers for INT8, dropping below 0.70 at layer 12.This indicates the relational geometry of the feature maps is severely getting distorted even before reaching the final layer which is the classification head.

2. Fairness & Performance by Quantization Precision

Figure 2 compares aggregate predictive performance against group-level fairness disparities across the three precision regimes.



(a) MPCs



(b) CFR

Figure 3: CFR & MCP

Precision	**Accuracy**	**Macro-F1**	**Mean EOD**	**Max |EOD|**
FP32	0.747	0.681	-0.008	0.507
FP16	0.747	0.681	-0.008	0.507
INT8	0.583	0.515	$+0.048$	0.366

Table 1: Performance and fairness on Bias in Bios (28 occupations, 99K test samples).

- **Macro F1-Score:** The FP16 Variant maintains performance parity with the FP32 baseline, but INT8 model shows a noticable F1-score drop, which is approximately 0.67 to 0.53.
- **Mean DPD & EOD:** Surprisingly, at this particular sample and threshold, the INT8 model has lower Demographic Parity Difference (DPD) and Equal Opportunity Difference (EOD) than the baseline model. This may happen because the large drop in performance of the INT8 model makes its predictions more uniform across different groups. However, this comes at the cost of lower overall accuracy and reliability of the model.

2. Bert Uncased on EEC

This is the same model trained using linear probing on the Jigsaw Unintended Bias dataset.

1.Mean Counterfactual Probability Shift (MCPS)

The MCPS serves as the primary indicator of demographic sensitivity, quantifying the average magnitude of variance in the model's predicted toxicity probability when a demographic identity term is swapped.

- **BaseLine:** FP32 and FP16 baselines showed a mathematically identical sensitivities. For both of these gender based MCPs is recodered as 0.045, where as the race based is recorded as 0.051.This indicates quantizing to FP16 doesn't harm much.
- **INT8 Representation Shift:** The 8-bit quantization yielded a significant drop. The gender one dropped to 0.036 and the race one dropped to 0.038.Here Instead of increasing demographic sensitivity,quantization has reduced the model's responsiveness to identity-related words because rounding removed small differences in the internal representations.

2.Counterfactual Flip Rate (CFR) Dynamics

The Counterfactual Flip Rate (CFR) measures the empirical fraction of test cases where a demographic substitution forces the predicted class $\hat{y}$ to completely flip across the binary decision boundary

1. Same like the MCPs even here, FP32 and FP16 models exhibited identical CFRs.
2. The transition to INT8 triggered a catastrophic collapse in the CFR The gender-based flip rate plummeted to near 0.1%, and the race-based flip rate reached absolute zero.

The reduction observed in the INT8 model may not necessarily indicate an improvement in fairness. Instead, it may be caused by the degradation of the model's internal representations due to quantization, which can also significantly harm predictive performance.

3. Linear Probing on Bias in Bios

Table 1 summarizes task performance and fairness metrics across the three precision levels.

Their is barely any loss of accuracy in the FP16 model (the F1 changes by less than 0.001). Whereas the INT8 models drops accruacy by 16.4 points and F1 by 16.6 points, this is a substantial drop from the baseline FP32 model.

For fairness metrics too FP16 is effectively lossless. INT8 howvveer, flpus the sign of mean EOD from -0.008 to $+0.048$, shifting th e model from a slight male advatage to a female advantge. As Figure

The representation analysis as seen in Figure 5 shows why this shift happens, INT8 CKA drops steadily from 0.97 at layer 1 to 0.77 at layer 12, L2 displacement grows from 0.72 to 7.54, and cosine similarity falls from 0.998 to 0.864. Each layer adds a small error that stacks up and makes the embedding drift to a different region.

The per-class EOD stats show that INT8 quantization does not degrade fairness in a uniform or predictable direction. Under the FP32 baseline, mean EOD sits at -0.008 which is very near the fairness line but individual occupations already carry substantial gender gaps (e.g. *model* at -0.507, *surgeon* at $+0.310$). After INT8 quantization, the aggregate mean EOD flips sign to $+0.048$, yet this number masks wildly divergent per-class behaviour visible in Figure 4: *rapper* swings by $+0.224$, *photographer* by $+0.160$, and *dentist* by $+0.164$, while *dietitian* shifts $+0.230$ in the opposite direction. No consistent pattern is seen in the shifts that can relate this phenomena to class size, baseline bias magnitude, or gender ratio in the training data. Here what we quantization introduces is not a systematic bias toward one gender but rather an increase in *fairness variance*, i.e. the model's per-class gender gaps become noisier and harder to predict. This can be more problematcic than a uniform shift, because a uniform shift can be detected and corrected with a single threshold adjustment, whereas occupation-specific shifts would be way harder to account for.

Jigsaw is a binary classification task, so quantization noise tends to push the model toward predicting one class more often, leading to a more uniform bias. In contrast, Bias in Bios is a 28-class classification task, where quantization noise affects each occupation's decision boundary differently, resulting in more varied and chaotic per-class behavior.

9 Timeline

Phase II – Generative Model Analysis

1. Architectural Expansion and Decoder-Only Baseline

The immediate next step involves selecting representative decoder-only architectures (e.g., GPT-2 or Llama-based models) to serve as the new experimental testbeds. Unlike the BERT-based approach that uses a frozen encoder for classification, this phase will evaluate the model's performance in zero-shot or few-shot generative settings. Initial efforts will focus on establishing full-precision (FP32) baselines for these models to ensure their generative outputs are stable and statistically representative before any quantization noise is introduced.

2. Extended Dataset Curation for Generative Tasks

As the project shifts to decoder-only models, we will identify and integrate new datasets specifically compatible with generative evaluation. While the Equity Evaluation Corpus (EEC) remains useful for counterfactual probing, we will search for and incorporate datasets designed for open-ended generation, such as BOLD (Bias in Open-ended Language Generation) or RealToxicityPrompts. These datasets will allow us to measure if quantization increases the likelihood of generating biased or stereotypical content, moving beyond discrete classification labels.

3. Quantization of Autoregressive Layers

We will apply post-training quantization, specifically INT8 dynamic range quantization, to the massive weight matrices within the self-attention and feed-forward blocks of the decoder. Given that decoder-only models accumulate errors autoregressively (where each generated token depends on previous noisy tokens), we will specifically monitor how quantization noise compounds during the generation of longer sequences.

4. Generative Fairness and Alignment Evaluation

The final stage of the timeline will focus on evaluating fairness through the lens of generation. We will compute the Counterfactual Flip Rate for generated text and utilize Centered Kernel Alignment (CKA) to map representational shifts across the decoder's layers. We aim to pinpoint whether the "asymmetric representational collapse" observed in encoder models—where minority group features are disproportionately distorted—is equally preva-

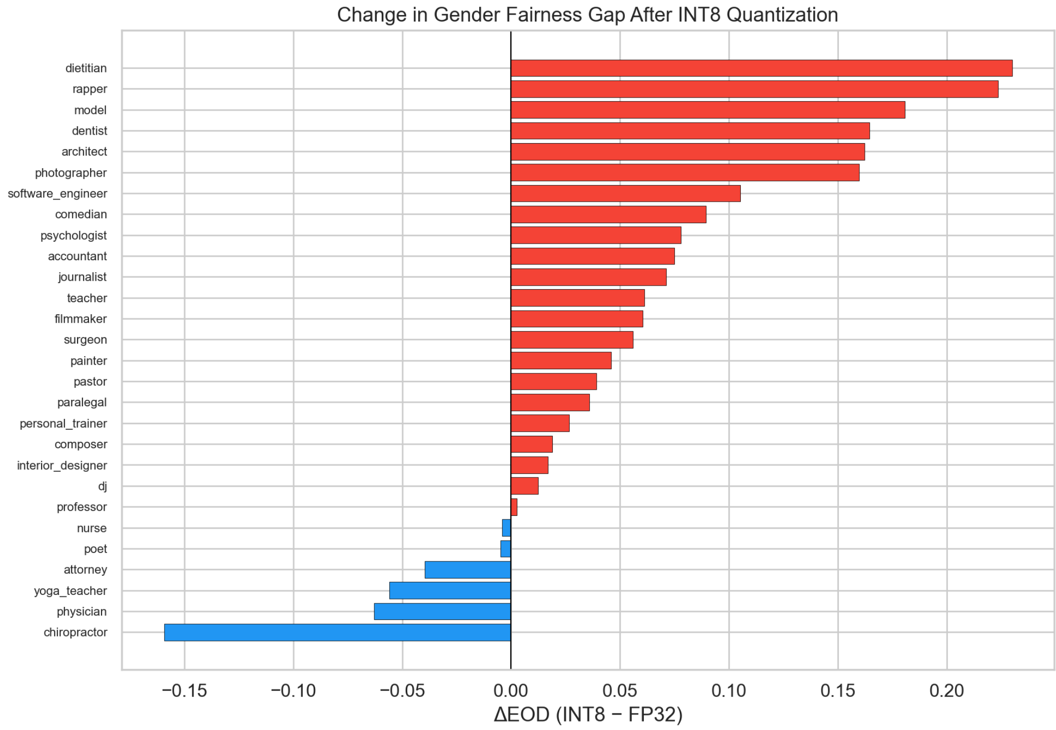


Figure 4: Per-occupation EOD shift after INT8 quantization ($\Delta\text{EOD} = \text{EOD}_{\text{INT8}} - \text{EOD}_{\text{FP32}}$).

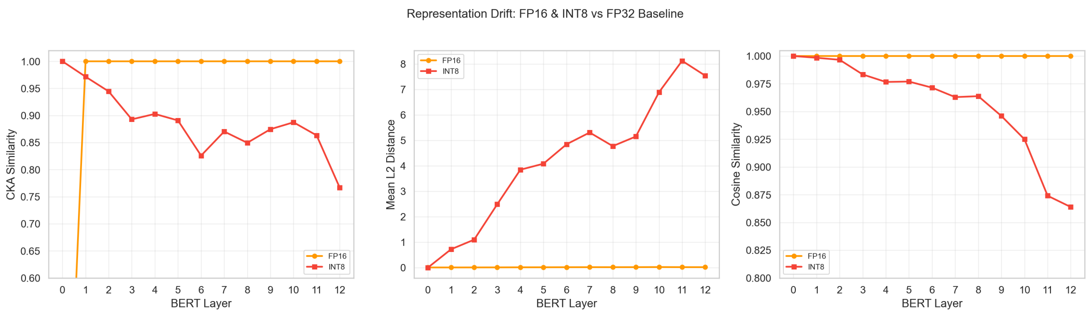


Figure 5: Layer-wise representation drift: CKA, L2 distance, and cosine similarity (FP16 & INT8 vs. FP32).

lent or further amplified in generative architectures.

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